Counterexamples to the proposed Redheffer lower bound for a ratio of normalized Bessel functions

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Abstract

The lower bound in MathDB 335189 fails at the explicit pair $(\nu, x) = (30, 1)$. We certify this with rational arithmetic and a variational bound for the first Bessel zero. More generally, for every real $\nu \geq 28$ it fails at all sufficiently small nonzero real x. The upper bound was already known in the literature. The mathematical counterexample is complete, but campaign eligibility remains pending inspection of the complete 2018 journal version of the source. No new research credit or priority is claimed here.

Task 1: the conjecture and a complete counterexample

For $\nu > -1$, define the even entire normalized Bessel function by

$$\mathcal{J}_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{4^k k!(\nu+1)_k}, \qquad \mathcal{J}_\nu(x) = 2^\nu \Gamma(\nu+1) x^{-\nu} J_\nu(x) \quad (x > 0).$$

Here $(a)_k = a(a+1)\cdots(a+k-1)$, with $(a)_0 = 1$. Write $j = j_{\nu,1}$ for the first positive zero of J_ν, $q = j^2$, and $R_\nu(x) = \mathcal{J}_{\nu+1}(x)/\mathcal{J}_\nu(x)$. Conjecture 1, equation (3.2), on page 8 of the current preprint [1] asserts, for $|x| < j$,

$$\left(\frac{q}{q-x^2}\right)^{q/[4(\nu+1)(\nu+2)]} \leq R_\nu(x) \leq \frac{q}{q-x^2}. \tag{1}$$

The source explicitly conjectures the left inequality for every $\nu > -1$; our examples satisfy that intended range.

Theorem 1. *For every real $\nu \geq 28$, there is $\varepsilon_\nu > 0$ such that the left inequality in (1) is strictly reversed whenever $0 < |x| < \varepsilon_\nu$. It is also strictly reversed at $\nu = 30,\ x = 1$.*

A bound for the first zero

Lemma 1. *For every real $\nu \geq 28$,*

$$j_{\nu,1}^2 \leq U_\nu := \frac{3(2\nu+7)(2\nu+8)}{10} < T_\nu := \frac{4(\nu+1)(\nu+2)(\nu+3)}{3\nu+5}.$$

Proof. Set $\phi(r) = J_\nu(jr)$, which is positive on $0 < r < 1$. The Bessel equation gives

$$-(r\phi')' + \frac{\nu^2}{r}\phi = j^2 r\phi.$$

For $u(r) = r^\nu(1-r)^3$, integration by parts gives the ground-state identity

$$\int_0^1 \left(r(u')^2 + \frac{\nu^2}{r}u^2 - j^2 r u^2 \right) dr = \int_0^1 r\phi^2 \left((u/\phi)' \right)^2 \, dr \geq 0. \tag{2}$$

For completeness, substitute $u = \phi v$ in the left side: its integrand is $r\phi^2(v')^2 + (r\phi\phi' v^2)'$. The boundary term $r\phi' u^2/\phi$ tends to zero at both ends. At zero, $\phi(r) = cr^\nu(1 + O(r^2))$; at one, $\phi(r)$ has a simple zero and $u(r) = O((1-r)^3)$. All the displayed integrals converge. Simplicity of the zero follows from uniqueness for the Bessel differential equation at $r = 1$.

Writing $u = r^\nu w$, integration of the cross term yields

$$\int_0^1 \left(r(u')^2 + \frac{\nu^2}{r}u^2 \right) dr = \int_0^1 r^{2\nu+1}(w')^2 \, dr, \qquad w = (1-r)^3.$$

Indeed, the removed boundary term is $\nu r^{2\nu} w^2$; it vanishes at both endpoints. Consequently (2) implies

$$j^2 \leq \frac{9\int_0^1 r^{2\nu+1}(1-r)^4 \, dr}{\int_0^1 r^{2\nu+1}(1-r)^6 \, dr} = 9\frac{B(2\nu+2,5)}{B(2\nu+2,7)} = U_\nu.$$

The beta-integral evaluation follows by repeated integration by parts. Finally, if $s = \nu - 28 \geq 0$, direct algebra gives

$$10(3\nu+5)(T_\nu - U_\nu) = 4s^3 + 246s^2 + 3854s + 2256 > 0.$$

□

Failure near zero for every order at least 28

Both normalized functions are positive near zero, so their real logarithms have convergent Taylor series there. The defining series gives

$$\log R_\nu(x) = \frac{x^2}{4(\nu+1)(\nu+2)} + \frac{(3\nu+5)x^4}{32(\nu+1)^2(\nu+2)^2(\nu+3)} + O(x^6).$$

The logarithm of the proposed lower bound is

$$\frac{q}{4(\nu+1)(\nu+2)} \log \frac{q}{q-x^2} = \frac{x^2}{4(\nu+1)(\nu+2)} + \frac{x^4}{8(\nu+1)(\nu+2)q} + O(x^6).$$

Thus their difference has coefficient

$$C_\nu = \frac{(3\nu+5)q - 4(\nu+1)(\nu+2)(\nu+3)}{32(\nu+1)^2(\nu+2)^2(\nu+3)q}$$

at x^4. Lemma 1 proves $C_\nu < 0$. For each fixed $\nu \geq 28$, analyticity therefore makes the difference strictly negative at every sufficiently small nonzero real x. We can choose $\varepsilon_\nu < j_{\nu,1}$, proving the first assertion. This is an analytic sign argument, not an inference from a finite set of Taylor coefficients or numerical samples.

An exact counterexample at $(30, 1)$

Set $q = j_{30,1}^2$, $U = 6834/5$, and $A = 3968 = 4 \cdot 31 \cdot 32$. Lemma 1 gives $q \le U$. Also $j_{30,1} > 1$: for $0 \le x \le 1$, the alternating defining series has decreasing term magnitudes and $\mathcal{J}_{30}(x) \ge 1 - x^2/124 > 0$. The claimed inequality is therefore defined at $x = 1$.

Let

$$S_{\nu,m} = \sum_{k=0}^{m} \frac{(-1)^k}{4^k k!(\nu+1)_k}.$$

The alternating-series remainder bound implies

$$0 < S_{30,5} \le \mathcal{J}_{30}(1), \qquad \mathcal{J}_{31}(1) \le S_{31,4}.$$

The exact partial sums are

$$S_{30,5} = \frac{4748440055099}{4786893619200}, \qquad S_{31,4} = \frac{7660705361}{7720796160}.$$

Consequently, with $d = 252139378/10^{12} > 0$,

$$R_{30}(1) \le \frac{4749637323820}{4748440055099} < 1 + d.$$

One exact check of the strict inequality is

$$d - \left(\frac{4749637323820}{4748440055099} - 1\right) = \frac{481473794211}{237422002754950000000000} > 0.$$

For $d \ge 0$, integration of $(1+t)^{-1} \le 1 - t + t^2$ gives $\log(1+d) \le d - d^2/2 + d^3/3$. On the other hand, since $1 < q \le U$,

$$\frac{q}{A} \log \frac{q}{q-1} = \frac{1}{A} \sum_{m=1}^{\infty} \frac{q^{1-m}}{m} \ge \frac{1}{A} + \frac{1}{2AU}.$$

The remaining rational comparison is

$$\frac{1}{A} + \frac{1}{2AU} - \left(d - \frac{d^2}{2} + \frac{d^3}{3}\right) = \frac{95976549656086382758355851863 79}{13240875000000000000000000000000000000000} > \frac{7}{10^{10}} > 0.$$

It follows that

$$\log R_{30}(1) < \frac{q}{A} \log \frac{q}{q-1},$$

which proves the explicit counterexample and completes the proof of Theorem 1. The accompanying verifier reproduces every rational comparison and the symbolic identities.

Scope of the conclusion and the known upper bound

A single admissible counterexample disproves (1) as a universal statement; the interval family supplies a second analytic proof. We do not assert failure at every order below 28. For context, the upper bound in (1) is already credited to earlier work by [1], page 8. Its short proof uses the classical Mittag–Leffler expansion and first Rayleigh sum recorded in that source:

$$R_\nu(x) = \sum_{k \ge 1} \frac{w_k}{1 - x^2/j_{\nu,k}^2}, \qquad w_k = \frac{4(\nu+1)}{j_{\nu,k}^2} > 0, \qquad \sum_{k \ge 1} w_k = 1.$$

Every denominator is at least $1 - x^2/j_{\nu,1}^2$ for real $|x| < j_{\nu,1}$, proving the upper inequality. Its exponent 1 is necessary because the first summand has a simple pole at $x = j_{\nu,1}$. This known upper bound earns no separate credit.

Task 2: determine the exact range of valid orders

Find the set of $\nu > -1$ for which the proposed lower bound holds throughout $(-j_{\nu,1}, j_{\nu,1})$. The inequality $j_{\nu,1}^2 \geq T_\nu$ is a necessary local condition. Determine whether it is sufficient and, if not, identify the additional obstruction. The proved $\nu = -1/2$ case in [1] provides a starting point.

Task 3: determine the true optimal lower exponent

Study

$$a_\nu = \inf_{0<x<j_{\nu,1}} \frac{\log R_\nu(x)}{\log(j_{\nu,1}^2/(j_{\nu,1}^2 - x^2))}.$$

Determine the minimizer and a useful formula or certified bounds. For $\nu \geq 28$, the proof shows that a_ν is strictly less than the proposed value $j_{\nu,1}^2/[4(\nu+1)(\nu+2)]$. Resolve its behavior as ν tends to infinity.

Task 4: classify the Rayleigh-coefficient monotonicity

The source proposes monotonicity in m of

$$\omega_{m,\nu} = j_{\nu,1}^{2m} \left(\sum_{k\geq 1} j_{\nu,k}^{-2m} - \sum_{k\geq 1} j_{\nu+1,k}^{-2m} \right).$$

The inequality $\omega_{2,\nu} \geq \omega_{1,\nu}$ is exactly the necessary condition $j_{\nu,1}^2 \geq T_\nu$. Classify the orders and indices where monotonicity changes, and determine whether the sequence has at most one change of direction.

Task 5: optimize elementary variational certificates

Replace $r^\nu(1-r)^3$ by $r^\nu(1-r)^p$ with $p > 1/2$. The same computation gives

$$j_{\nu,1}^2 \leq \frac{p(2\nu+2p+1)(2\nu+2p+2)}{2(2p-1)}.$$

Optimize this bound, or extend the trial space by a polynomial factor, to obtain the smallest provable threshold for counterexamples using elementary exact inequalities.

References

[1] Á. Baricz and K. Mehrez, *Redheffer type bounds for Bessel and modified Bessel functions of the first kind*, arXiv:1701.08446v1 (2017), 9 pp., `https://arxiv.org/abs/1701.08446`. Complete preprint inspected; Conjecture 1 is on p. 8. Published in *Aequationes Mathematicae* 92 (2018), 425–439, `https://doi.org/10.1007/s00010-018-0545-8`. The complete journal text has not been inspected.